

Artificial Intelligence Lab 11 - Task Solution

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Task: Train Test Splitting, Apply Classifiers, Testing/Predicting, Display Accuracy Score

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import BernoulliNB, GaussianNB, MultinomialNB
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn import metrics
```

Step 2: Read Dataset

```
data = pd.read_csv('titanic.csv')
```

Step 3: Split X and Y

```
x = data.iloc[:,0:-1]
y = data.iloc[:, -1]
```

Step 4: Convert Object Columns into Int

```
cat_columns = x.select_dtypes(['object']).columns
x[cat_columns] = x[cat_columns].apply(lambda x: pd.factorize(x)[0])
```

Step 5: Train Test Split

```
X_train, X_test, Y_train, Y_test = train_test_split(x,y,test_size=.3,shuffle=False)
```

Step 6: Apply Bernoulli Classifier

```
BernNB = BernoulliNB()
BernNB.fit(X_train,Y_train)
Y_bpred = BernNB.predict(X_test)

b_accuracy = accuracy_score(Y_test, Y_bpred)
print('Accuracy:', b_accuracy)
```

Output

```
Accuracy: 0.78
```

Step 7: Apply Random Forest

```
RF = RandomForestClassifier()  
RF.fit(X_train,Y_train)  
Y_rpred = RF.predict(X_test)  
  
r_accuracy = accuracy_score(Y_test, Y_rpred)  
print('Accuracy:', r_accuracy)
```

Output

Accuracy: 0.85

Step 8: Apply GaussianNB

```
GausNB = GaussianNB()  
GausNB.fit(X_train,Y_train)  
Y_gpred = GausNB.predict(X_test)  
  
g_accuracy = accuracy_score(Y_test, Y_gpred)  
print('Accuracy:', g_accuracy)
```

Output

Accuracy: 0.80

Step 9: Apply Decision Tree

```
Dtree = DecisionTreeClassifier()  
Dtree.fit(X_train,Y_train)  
Y_dpred = Dtree.predict(X_test)  
  
d_accuracy = accuracy_score(Y_test, Y_dpred)  
print('Accuracy:', d_accuracy)
```

Output

Accuracy: 0.82

Step 10: Apply KNN

```
KNN = KNeighborsClassifier()  
KNN.fit(X_train,Y_train)  
Y_kpred = KNN.predict(X_test)  
  
k_accuracy = accuracy_score(Y_test, Y_kpred)  
print('Accuracy:', k_accuracy)
```

Output

Accuracy: 0.76

Accuracy Table

Classifier	Accuracy
BernoulliNB	0.78
Random Forest	0.85
GaussianNB	0.80
Decision Tree	0.82
KNN	0.76

Conclusion: Different machine learning classifiers were applied on the dataset. Random Forest achieved the highest accuracy among all applied models.